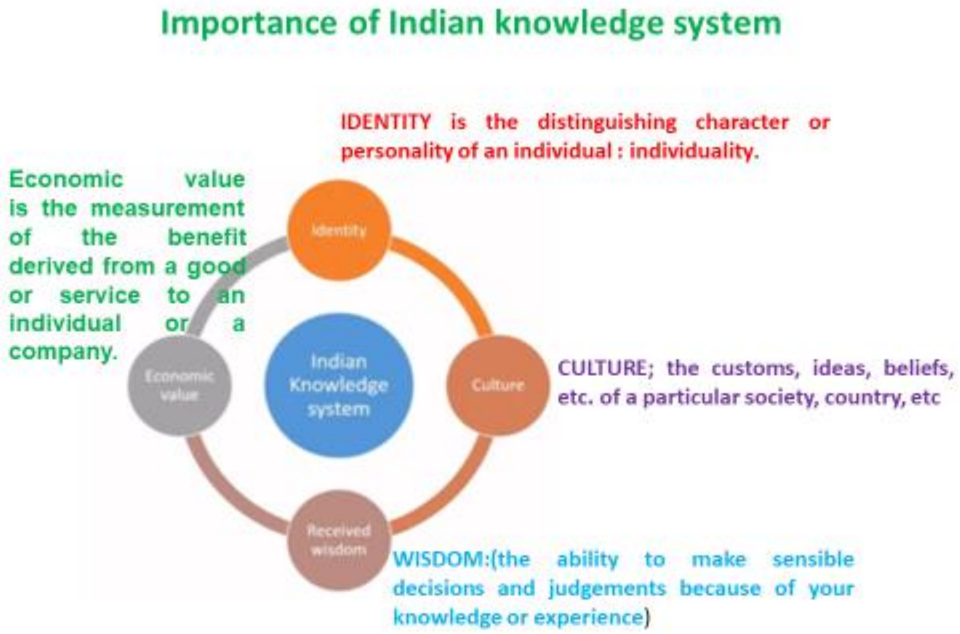
IKS UNIT – I

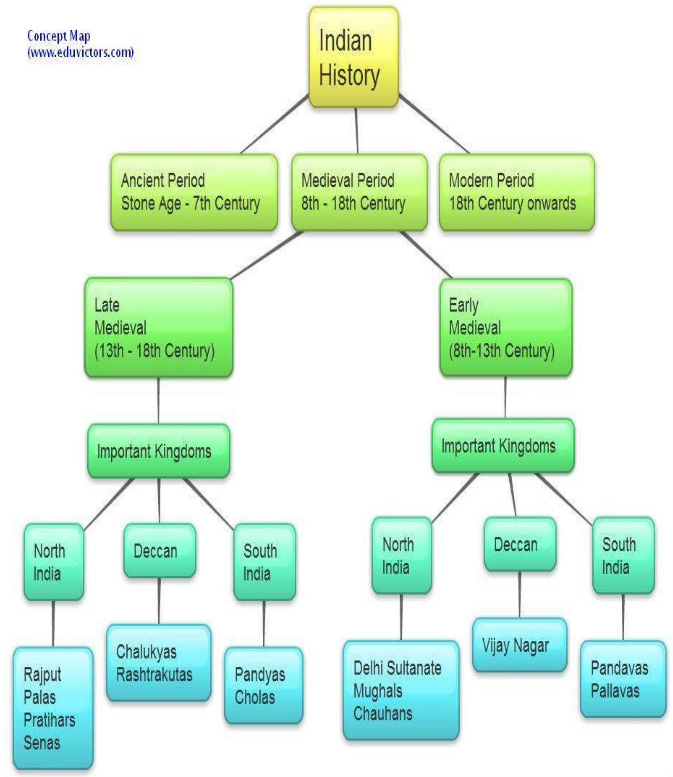
IKS DEFINITION (INDIAN KNOWLEDGE SYSTEM)

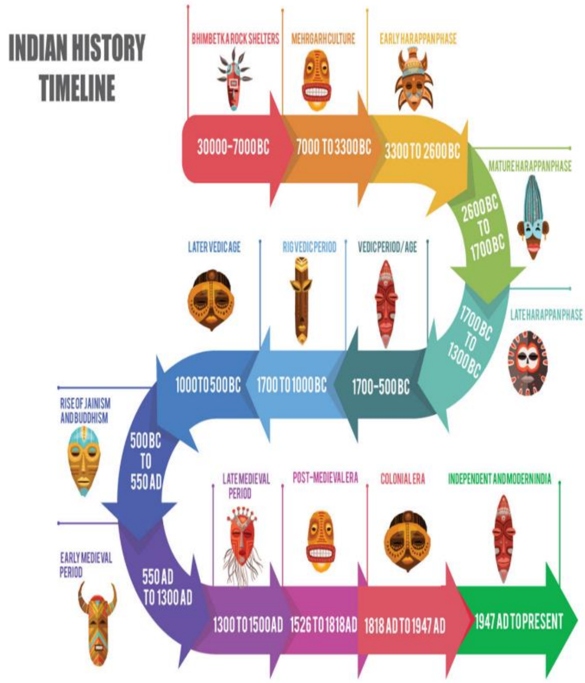
Systematic, well-structured system and process of knowledge transfer from generation to generation.
The term 'Indian Knowledge System' (IKS) refers to the cumulative knowledge, wisdom, and traditions that have developed within the Indian subcontinent over centuries.
Breaking it down:
"Indian" denotes the geographical and cultural context of the knowledge, highlighting its origin within the undivided Indian subcontinent.
"Knowledge" encompasses a wide range of domains, including science, philosophy, arts and more.
"System" suggests that IKS is an organized, structured, and interconnected body of knowledge that is comprehensive and coherent



1. **Preservation and Transmission of Knowledge**: To preserve traditional knowledge systems and transmit them from one generation to the next, ensuring continuity and cultural richness.
2. **Harmony with Nature**: To promote an understanding of the interconnectedness of all living beings and the importance of living in harmony with nature, as reflected in various aspects of Indian philosophy.
3. **Social Harmony and Ethics**: To instill values of social justice, compassion, and ethical conduct in individuals, contributing to the overall well-being of society.
4. **Innovation and Adaptability**: To encourage innovation while staying rooted in traditional wisdom, fostering a balance between continuity and adaptation to changing circumstances.
5. **Identity**
6. **Culture:** set of ideas, customs. To shape the culture of society
7. **Received wisdom:(**the ability to make sensible decisions and judgements because of your knowledge or experience) provide path to transmit the innovations to next generation
8. **Economic value**: patent
9. **Cultural Heritage**: Ancient knowledge systems preserve cultural traditions, values, and practices, which are essential for maintaining cultural identity and continuity.
10. **Sustainability**: Many traditional knowledge systems offer sustainable solutions to modern problems, particularly in the realms of agriculture and environmental conservation.
11. **Holistic Well-being**: Systems like Ayurveda and Yoga provide comprehensive approaches to health and well-being, addressing both physical and mental aspects.
12. **Diversity and Innovation**: Ancient knowledge offers alternative perspectives and solutions, fostering diversity and innovation in addressing contemporary challenges.

HISTORICAL PERSPECTIVE OF INDIAN CIVILIZATION 3000BCE TO 2000CE





Periods of Indian Civilization

The development of Indian Civilization can be divided in seven periods

1.Original Indians (1700BCE-3300 BCE) Indus Valley Civilizations: Mohenja-daro and Harappa People
2. Aryans (2500BCE-322BCE) India's Root Culture
3. The Mauryan Empire (322BCE-188BCE) Spread of Buddhism
4. Gupta Period (320CE-480CE) Golden Age of India
5. Mugal Period (1175CE-1800CE) Turks and Mughals
6. European Rule (1800CE-1947CE) Portuguese, French, Dutch and English
7. 1947 Independence of India

Indus Valley Civilization . Indus Valley Civilization (1700BCE-3300BCE) Excavation in Northwest part of Pakistan has discovered this civilization that is over 5,000 years old. This is the period where people in the rest of the world were nomadic. **Aryans** (2500BEC-322BEC) Vedic Aryan civilization was a rural not an urban civilization and raw material remains from it are extant; hence little in the form of archeological evidence. No written documents that record this invasion. In their absence: the Vedas, a collective term for the ancient wisdom preserved in texts about rituals, priests, and speculations about the nature of the human and divine worlds and transmitted orally.

Maurya Empire (322 BCE-185 BCE) Chandragupta Maurya began in 322 BC establishing a great empire in northern India and the lands abandoned by Alexander the Great. The mauryan Empire included all of present-day northern India and much of modern Afghanistan. The Mauryans were better rulers and culturally rich. According to legend, Chandragupta retired from the throne after ruling for twenty-four years, passed it to his son, and became a monk and starved himself to death. After that Ashoka converted to Buddhism and introduced Buddhism outside India.

Gupta Period (320 CE- 480 CE) After the decline of mauryans, Gupta rulers rose to power and over time, they came to rule much of India north of the Deccan Plain. The Gupta Period in Indian history is known as the Golden Age of India. It is the Era of the most advanced civilization, flush with wealth, higher education, trade with foreign countries, and an overall happy life. Here we found religious tolerance and freedom of worship.

Mugal Period- Turks and Mughals (1175 CE– 1800 CE) Turks from Central Asia invaded India and ruled from 1175 CE to 1340 CE. Attracted by India's wealth, looted and destroyed temples. More interested in wealth rather than .Turks' dominance ended in 1526 with the invasion of Mughals from central Asia.

European Rule (1800-1947) – Discovery of India The Invasion of Alexander had boosted trade contact outside India. Columbus, in his quest to find India ended up in North America and erroneously thought he had reached India, calling the native of the new land as Indians., the Portuguese fought and established their dominance and appointed Portuguese Governor in India.

British East India Company (1600 CE – 1858 CE)

Founded on December 31, 1600, the British East India Company marked the beginning of British trade dominance in India. The company's policies, including the Doctrine of Lapse and the introduction of the Enfield rifle, led to widespread discontent among Indians.

Independence. The main Historical Figures of the India Independence movement include Mahatma Gandhi, Jawaharlal Nehru, and Mohammad Ali Jinnah. India and Pakistan become free and independent countries on August 15, 1947. Nehru became the first prime minister of newly formed democratic country of India. Muhammad Ali Jinnah became the first governor general of Pakistan.

India as a Republic: 1950 CE – Present

As a republic, India's diverseculture, democracy, and unity emerged as definingfeatures. The nation's commitment to secularism, as enshrined in the Constitution, has played a significant role in preservingreligious and cultural diversity.Indian Constitution enshrines principles of liberty, equality, and justice.Marks India's transformation into a democratic republic.Symbolizes the struggle for freedom and the painful division during partition.Post-independence: India undergoes economic reforms and growth.

EDUCATION SYSTEM IN ANCIENT INDIA

According to the eras, Our education is divided into different system:

1.Ancient Education or Gurukul Education System in Vedic era (3000 B.C.E)and Buddhist era(500 B.C. to 1200 CE)

2. Medieval Education System Or Islamic Education System in Muslim era(1200 CE. to 1700 CE)

3.British Education System Or Pre-Independence Education System in British era(1800CE. to 1947 CE)

4.Post Independence Education System(After 1947)

5. Modern Education System

Figure 2: Classifications of Indian Knowledge Systems

1.Ancient education

During the ancient period, two systems of education were developed, Vedic, and Buddhist. The medium of language during the Vedic system was Sanskrit,while those in the Buddhist system were pali.

During those times the education was of Vedas, Brahmanas, Upnishads, and Dharmasutras. From the Rigveda onwards, our ancient education started with the objective of developing the students not only in the outer body but also on the inner body.

The ancient education focused on imparting ethics like humility, truthfulness, discipline, self-reliance, and respecting all creations to the students. The education was mostly imparted in ashrams, gurukuls, temples, houses. Sometimes pujaris of the temples used to teach students

2. Medieval education

During the eighth century Anno Domini (A.D) a huge number of Mohammadian invaded India.The Arabs and the Turks bought some new cultures, traditions, and institutions in India, in that the most remarkable change was the Islamic pattern of education which was different from the Buddhist and Brahmanic education system.The medieval age, education system primarily focused on the Islamic and Mughal System.

3. British education

In the 1830s Lord Thomas Babington Macaulay introduced the English language.The subjects and the syllabus were limited to some extent,

4.Post-Independence

After gaining independence in 1947, India's education history took a significant leap forward.The government implemented policies aimed at increasing access to education for all, emphasizing the importance of literacy and elementary education. The establishment of premier institutions like the Indian Institutes of Technology (IlTs) and Indian Institutes of Management (IIMs) underscored India's commitment to fostering excellence in higher education.

5.Modern

As India entered the 21st century, the history of education continued to evolve. As time passed education started to develop and entered into the modern era that is in the twenty-first century, the era of science, technology, and innovations. demand increases our education sector also needs to change and adapt to that environment

The nation confronted challenges related to quality, accessibility, and equity in education. Yet, advancements in technology and the rise of e-learning platforms opened new horizons for learning. Online education, digital classrooms, and skill development programs have revolutionized knowledge dissemination, ushering in a new era of education in India.

Education system

1. **Gurukula system**
2. **Vishwavidhayalas; Ancient Indian Universities**

GurukulSystemis an ancient Indian concept of education, wherein the participants get knowledge, by residing with his teacher as part of his family.(Residential in Nature), Subjects taught;- Śikshā (Phonetics) ,Vyākarana (Grammar),Jyotisha (Astronomy) ,Arthaśāstra (Economics),Dharmaśāstra (Laws) ,Śastravidyā (Art of Warfare) ,Kalā (Fine Arts),Mathematics, ,fundamentals of Astronomy, ,Science, Languages,, Medicinal theory

Generally it was the Brahmin households that used to run Gurukuls. They were situated inside and outside the villages or towns. There was never a barrier of rich or poor in a Gurukul the doors were always open to various deserving students.

Vishwavidhayalas; Ancient Indian Universities

Major universities was

1. Takshashila University
2. Nalanda University
3. Vikramshila University
4. Valabhi University
5. SomapuraMahavihara
6. Kanchipuram University
7. Pushpagiri University
8. SharadaPeethansta
9. JagaddalaMahavihara
10. OdantapuriMahavihara

Nalanda University

The word **NALANDA** derived from ‘nalam’ means lotus, (the symbol of knowledge) and ‘da’ means to give, so Nalanda means “Giver of Knowledge”.

Location;- city of Rajagriha (now Rajgir) and about 90 kilometers (56 mi) southeast of Pataliputra now Patna). District Nalanda, Bihar

The first residential university of the world.Nalandaan an ancient centre for higher education .It was founded by the Kumaragupta (Gupta Empire,)in 427CE and flourished for over 800 years till the end in 12th century CE

It accommodated over 2,000 teachers and 10,000 students from all over the world (Korea, Japan, China, Tibet, Indonesia, Persia, and Turkey) .

It was devoted to Buddihist studies but also trained in

Courses taught in the University :

1. Political Science,
2. Administration,
3. Medicine,
4. Astronomy,
5. Mathematics,
6. Agriculture,
7. Trade,
8. Military Training,
9. Weapons Manufacturing,
10. Fine Arts,
11. Economics,
12. Philosophy,
13. Law

University is destroyed three times by kings at that times and also re- built twice by king Harshavardhan . But real destruction of nalanda university is done by Mohammad BakhtiyarKhilji in 1197.Efforts are being made to re-establish the Nalanda University by the Govt. of India

Takshashila University

Takshashila was in the kingdom of Gandhar, in Ancient India before partition.But now Takshashila is in Rawalpindi district of the Punjab Pakistan
It had 10,500 students including those from Babylon, Greece, Syria, and China. 68 subjects were taught at this university and the minimum entry age, ancient texts show, was 16.A wide range of subjects were taught by experienced masters:
It is thought to have existed around the 5th century BC.
At this location, Chanakya is said to have written the Arthashastra.
Theologies of both Buddhism and Hinduism were taught here.
Takshashila's notable teachers and students include Chanakya, Charaka, Panini, Jivaka, Prasenajit, and others.

1. Vedas,
2. Language,
3. Grammar,
4. Philosophy,
5. Medicine,
6. Surgery,
7. Archery,
8. Politics,
9. Warfare,
10. Astronomy,
11. Astrology,
12. Accounts,
13. Commerce,
14. Futurology,(study of the future)
15. Documentation,
16. Occult, (magical powers and activities)
17. Music,
18. Dance,

Vikramashila

It is a Buddhist tantric centre of learning, situated in **Kahalgao, District-** Bhagalpur, Bihar, India
It was founded by King Dharmapala between the late eighth and early ninth century.
It is said that the Vikramshila campus had six different buildings each belonging to a single stream of education, very much like we have different buildings for different departments now.The rage included topics like Sanskrit grammar, logic, metaphysics, philosophy, Buddhism, Tantra, and ritualism
Vikramshila University drew students from all over the country as well as from other countries. Over 1000 students were taught by over 100 teachers.
Subjects were taught here:

1. Political Science,
2. Administration,
3. Medicine,
4. Astronomy,
5. Mathematics,
6. Agriculture,
7. Trade,
8. Military Training,
9. Weapons Manufacturing,
10. Fine Arts,
11. Economics,
12. Philosophy,
13. Law

Valabhi University

Valabhi University is an ancient learning institution, located in modern-day Bhavnagar, Gujarat.
Valabhi University was a major Buddhist study center between 600 and 1400 CE.
This university used to receive royal funds for its sustenance and it was the lineage of the Maitrak dynasty, who later on became the patron of this university and designed its infrastructure further.
By the mid-Seventh century CE, Vallabhi had gained popularity for teaching Buddhist philosophy and Vedic literature.
It had more than 6000 students and 400 teachers.

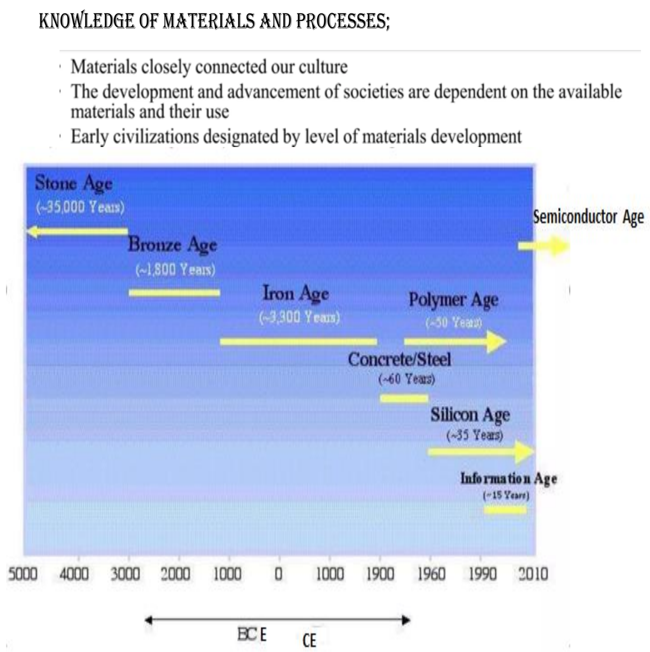
Some of the popular subjects taught at this university were Economics, Statesmanship, Business, Book-keeping and Agriculture.
Aside from religious sciences, the following courses are available:

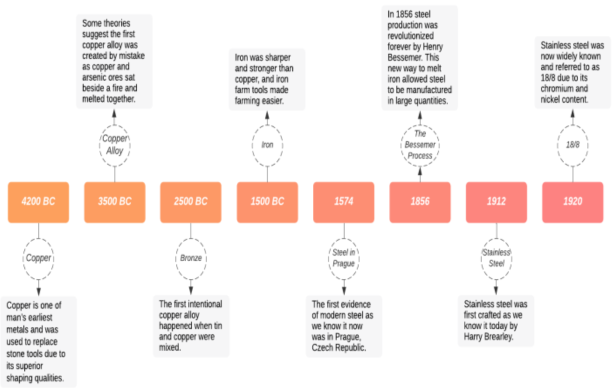
1. Niti (Political Science, Statesmanship)
2. Varta (Business, Agriculture)
3. Administration
4. Philosophy and Religious Thought (especially Buddhist philosophy)
5. Accounting, Economics, and Law

This university continued to function normally till the mid-Twelth century and was later destroyed by the Muslim invaders.
The foreign attacks had weakened the influence of local kings and it also disrupted the financial support this institution used to receive.

KNOWLEDGE OF MATERIALS AND PROCESSES;

The first evidence of metal in the Indian subcontinent - Mehrgarh in Baluchistan - small copper bead dated 6000 BCE
Mankind had knowledge of metals, alloys & its properties 4000 yrs ago .Different metals and alloys that were in use 3000 years ago
Indians were quiet advance in the use of Bronze, brass, Iron, Gold, silver, steel, Zinc, Tin, lead ,mercury and other alloy
Indians were first to introduce Zinc to human being and also to developed Cu-Zn alloy
88wt.% Copper + 12 wt.% Tin = Bronze
66% Copper + 34 wt.% Zinc = Brass
75% Gold + balance of copper, silver is added
Iron and its alloys
In India most metal idols which are placed in the temple are made out of : PANCHA LOHA – (Cu-Ag-Au-Zn-Sn)





- **Gold and silver;-**1st millennium BCE.
- 4200 – 2500 BCE Copper - Bronze
- 1500 – 300 BCE Iron
- 0 –cement
- Zinc ;-4th to 3rd century BCE.
- Wootz and steel;300 BCE
- 1574 – 1900 Steel
- 1900 CE- polymers
- 1912 –Stainless Steel
- Silicon;-1990
- 2000CE- nanomaterials and nano composites
- 2000 CE-2010 CE;- birth of high entropy alloy

- 2010 composite material

Metal used in ancient India

Classification of metals in Ancient Indian literature

According to ancient Indian literary reference, in nature there are four

1. suddhalohas (nativemetals) viz., suvarṇa (gold),
2. rajata (silver),
3. tamra (copper) and
4. loha (iron).

In addition there aretwo putilohas- naga (lead) and vaṅga (tin) and three misraloha (alloys) viz., pittala (brass),kaṃsya (bell metal) and vartaloha (an alloy made from fiv metal)

1. Bronze,
2. brass,
3. Iron,
4. Gold,
5. silver,
6. steel,
7. Zinc,
8. Tin,
9. lead ,
10. mercury
11. **arsenic,**
12. **antimony,**
13. other alloy

- 88wt.% Copper + 12 wt.% Tin = Bronze
- 66% Copper + 34 wt.% Zinc = Brass
- 75% Gold + balance of copper, silver is added
- Iron and its alloys
- PANCHA LOHA – (Cu-Ag-Au-Zn-Sn

use of metal

Ancient Indian have made use of base metals and alloys in multiple ways

1. Idols,
2. Utensils, vessels
3. Ornaments
4. Stamp seals
5. For the purposes of trade coins of Gold, Silver& Copper were in use.
6. Recorded history is 300-400 BCE However it could have been even earlier. Coins have inscriptions of the respective dynasties
7. Military aids (weapons)
8. Tools
9. Artefacts
10. Medical appliances
11. House and buildings
12. Ship building

Material/ metal processing

Ancient Indians contributions to metal working , metallurgy and metal science are noteworthy.Indian metalsmiths had notable achievements particularly developing unique skill of mining and working with iron , Steel , copper , Zinc and its alloys

India was the world leader in metallurgy for more than 12,000 years.

Copper and Gold : Melting point is 1100 deg.C

Silver960 deg.C

Lead (Pb) = 330 deg.C

- Tin (Sn) = 230 deg.C
- Zinc (Zn) = 410 deg.C

In-depth knowledge of ore-extraction, refining, alloying of copper, iron, tin, silver, gold, zinc, lead. In-depth knowledge of iron alloys which are resistant to corrosion and wear

The smelters & metalsmiths had gained high degree of knowledge regarding

- Furnace & furnace design
- Operation of furnace at desired temperature
- Combustion of fuel
- Refractories
- Temperature specification and other conditions for metallurgical process
- Alloy technology and Production of alloy of control composition
- .Designing and casting a variety of artefacts , moulding and die-casting

List the important metal working process that was in practice

a) Casting
b) Forging, Stamping (1200 BCE)
c) Ore-extraction and Ore-refining
d) Furnace and refractory's
e) Lost Wax Method,
f) Furnace
g) Refractories & Crucibles
h) Fuel
i) Temperature measurement

MATHEMATICS;

- Mathematics, referred to as Ganita is an integral part of Indians from very ancient times.
- Ancient Indians developed several concepts of mathematics primarily in the course of solving several real-life problems that they were facing.
- For example, one of the Veclangas is Jyotip, which confines itself with required mathematical concepts, measurements, and approximation techniques to make certain predictions of the movement of celestial bodies in the sky. Similarly,
- Veclairga, Kalpa required principles of geometry to construct Vedic altars of many different shapes.
- Buddhist and Jain were also deeply interested in mathematical concepts and produced canonical works discussing some useful mathematical concepts.
- What started as a requirement at an early stage continued through the post-Vedic period as an uninterrupted tradition and many useful contributions to the field of mathematics were made throughout Indian history.
- Over the years, the contributions spanned almost all modern areas of mathematics. These include algebra, geometry, trigonometry, indeterminate equations, square, cube, and their roots, permutation and combinations, numerical methods and approximations, infinite series, to name a few.

History of Mathematics in India

The history of Indian mathematics dates back to the vedic period (around 1500 B.C.) The 'Sulbasutras' of this vedic age are texts on rules for altar construction. They are the oldest texts on Indian mathematics. They contain the general enunciation of the Pythagoras theorem, approximate value for square root of 2, methods of transforming one figure to another etc... Indians also used the decimal place-value system of representing the numbers. They had a representation for zero. The origin of the decimal place-value system in India was sometime around 1st century B.C.

Mathematics had played a significant role in the development of Indian culture and tradition. Ancient Mathematics in India or the Vedic Mathematics had begun in the early Iron Age. The Shatpata Brahmana and the Sulabasutras included things like irrational numbers, prime numbers, rule of three, and square roots and also solved many of the complex problems.

During the Indus valley civilization the sites of Harappa and Mohenjo-Daro constructions took place following precise mathematical calculation. The measurement that was used in the construction was decimal in nature and at this juncture it can be said that mathematics in the ancient age was used for the purpose of construction. Many of the Vedic texts give details about some of the geometric constructions which were used during the Vedic Age. Besides the Vedic texts and the Sulabasutras also gave enough details about the geometric constructions those were used in the ancient age.. Although during this period mathematics was mostly used for solving practical problems but at the same time there was a little development in the field of algebra.

- Vedic
- Classical
- Medieval to Mughal period

- Born in 1800s
- Born in 1900s

use of mathematics

1. solving several real-life problems
2. Astronomy;- predictions of the movement of celestial bodies in the sky
3. Taxation
4. Commerce
5. Trade
6. Calendar
7. Structure
8. Jyotisha'
9. Surveying
10. Construction

contribution of Indian mathematicians

There is doubt that the world today is greatlyindebted to the contributions made by IndianMathematicians. One of the most importantcontribution made by Indian Mathematicians ARE

1. introduction of decimal system
2. invention of zero.
3. The number system
4. The zero
5. Vedic Mathematics and arithmetical operations
6. Geometry of the Sulba Sutras
7. Classical contribution to Indeterminate Equations and
8. Algebra
9. Indian Trigonometry
10. Kerala contribution to Infinite Series and Calculus.
11. Pithagoras theorem

AIRTHEMATIC

- Square of number
- Square root
- Square root of imperfect system
- Series and progression

GEOMETRY

- Property of right triangle
- The value of $\prod$

Trigonometry

Algebra

Binary mathematics and combinatorial probles

Magic square

Great mathematicians and their contribution

Famous Indian Mathematicians and their Inventions

Here are some of the great mathematicians of India and their inventions:

Mathematicians	Renowned Works and Inventions
Aryabhata	Formula: (a + b) 2 = a2 + b2 + 2ab
Brahmagupta	Introduction of zero (0)
Srinivasa Ramanujan	Properties of the Partition Function
P.C. Mahalanobis	Mahalanobis Distance
C.R. Rao	Theory of Estimation
D.R. Kaprekar	Devlali numbers, Kaprekar numbers, the Harshad numbers and Demlo numbers.
Harish Chandra	Representation theory, Harmonic analysis on semisimple Lie groups.
Satyendra Nath Bose	Collaboration with Albert Einstein, Modern theoretical physics in India.
Bhaskara	Declared that any number divided by zero is infinity and that the sum of any number and infinity is also infinity.
Narendra Karmarkar	Karmarkar's algorithm

MATHEMATICS

TABLE 8.2 Ancient Indian Mathematicians and Their Salient Contributions

Sl. No.	*Details of the Work/Mathematician*	*Period, Location*	*Salient Contributions*
1	**Vedic Texts**	3000 BCE or earlier	The earliest recorded mathematical knowledge, number system, Pythagorean type triplets; Decimal system of naming numbers, the concept of infinity;
2	**Lagadha** – Vedāṅga-jyotiṣa	~ 1300 BCE	Astronomical concepts; a mathematical model for sun-moon movement in time; equinoxes and solstices;
3	**Śulba-sūtras** (Baudhāyana, Āpasthamba, Kātyāyana and Mānava Śulba-sūtras)	800–600 BCE	Earliest Texts of Geometry; Approximate value of the square root of 2, and π. Exact procedures for the construction and transformations of squares, rectangles, trapezia, etc.
4	**Pāṇini** – *Aṣṭādhyāyī*	500 BCE; Śalātura (in Khyber province in Pakistan)	Algorithmic approaches; Originator of the Backus–Naur Form (BNF), used in programming languages today, Context-sensitive rules, Arrays, inheritance, polymorphism;
5	**Piṅgala** – *Chandaḥ-śāstra*	300 BCE	Binary sequences; Conversion of Binary to Decimal system and vice versa; 'Meru Prastara' (Pascal's triangle); Optimal Algorithms to calculate powers; Zero as a Symbol;
6	**Bauddha Mathematical Works**	about 500 BCE to 500 CE	Multi-valued logic, Discussion of indeterminate and infinite numbers;
7	**Jaina Mathematical Works** – *Sūrya-Prajñapti, Jambūdvipa-prajñapti, Bhagavatī-sūtra, Sthānāṅga-sūtra, Uttarādhyayana-sūtra, Tiloyapannati, Anuyoga-dvāra-sūtra*	200 BCE to 300 CE	Concept of logarithms, large numbers; algorithms for raising a number by a power; the arc of a circle; combinatorics; mensuration; Decimal system; Approximation of π;
8	**Āryabhaṭa** – *Āryabhaṭīyam*	476–550 CE; Kusumapura, near Pataliputra, Bihar	Concise verses; Algorithm for square root, cube root, Place value system; Sine table; geometry; quadratic equations; Linear indeterminate equations; Sums of squares and cubes of numbers; Planetary astronomy; Plane and spherical trigonometry;
9	**Varāha Mihira** – *Bṛhat Samhitā, Bṛhat-jātaka, Pañca-siddhāntikā*	482–565 CE, Ujjain, Madhya Pradesh	Summary of five ancient siddhāntas; Sine table, trigonometric identities; ($\sin^2 + \cos^2$); combinatorics; Magic squares;
10	**Bhāskara I** – *Commentary on Āryabhaṭīya, Laghu-bhāskarīyam and Mahā-bhāskarīyam*	600–680 CE; Vallabhi region, Saurashtra, Gujarat	Expanded Āryabhaṭa's work on Integer solution for indeterminate equations; Approximate formula for the sine function, Planetary Astronomy;

Concept of Infinity, Decimal

Trigonometry

Binary System 0, 1

Algebra

Sl. No.	Details of the Work/Mathematician	Period, Location	Salient Contributions
11	**Brahmagupta** – *Brahmasphuṭa-siddhānta; Khaṇḍakhādyaka*	598–668 CE; Bhillamala in Rajasthan	Rules of arithmetic operations with zero and negative numbers, Algebra (Bijagaṇita); linear and quadratic indeterminate equations; Pythagorean triplets, Formula for the diagonals and area of a cyclic quadrilateral; notion of arithmetic mean.
12	**Virahāṅka** – *Vṛttajātisamuccaya (in Prākṛt)*	~600 CE	Fibonacci numbers; Moric metres.
13	**Śrīdharācārya** – *Triśatikā and Pāṭīgaṇita*	870–930 CE; Bhūrisṛṣṭi or Bhurshut village, Hugli, West Bengal	Arithmetic, Algebra, and Commercial Mathematics; Approximation of square root of a non-square number; Quadratic equations; Practical applications of algebra;
14	**Mahāvīrācārya** – *Gaṇita-sāra-saṅgraha*	800–870 CE; Gulbarga Karnataka;	A comprehensive, exclusive textbook on mathematics covering arithmetic-geometry- algebra. Continuing the ancient Jaina mathematics tradition; permutations and combinations; arithmetic and geometric series; the sum of squares and cubes of numbers in arithmetic progression;
15	**Jayadeva**	10th Century CE or earlier	Cakravāla (cyclic) method for solution of the second-order indeterminate equation.
16	**Shripati** – *Gaṇita-tilaka, Siddhānta-śekhara, Dhīkoṭidākaraṇa, etc.*	1019–1066 CE; Rohiṇīkhaṇḍa, Maharashtra	Planetary Astronomy
17	**Bhāskarācārya (Bhaskara-II)** – *Līlāvatī on arithmetic and geometry; Bījagaṇita on algebra; Siddhānta-śiromaṇi on astronomy; Vāsanābhāṣya on Siddhānta-śiromaṇi.*	1114–1185 CE; Hailed from Bijjadavida	Canonical textbooks used all over India, Detailed explanations including Upapatti (demonstration or proof); addition formula for sine function. Surds; permutations, and combinations; Solution of indeterminate equations, Ideas of calculus, including mean value theorem, planetary astronomy; construction of several instruments;
18	**Nārāyaṇa Paṇḍita** – *Gaṇita-kaumudī – a treatise on arithmetic and Bījagaṇita-aātāṁśa – a treatise on algebra.*	1325–1400 CE;	Advanced textbooks taking forward the works of Bhāskarācārya, further properties of cyclical quadrilaterals, summation and repeated summations of arithmetic series, theory and construction of Magic squares, further developments in combinatorics.

ARYABHATA (476—550AD)

Aryabhatta was born in 476 A.D. Kusumpur, India. He was the first in the line of great mathematicians from the classical age of Indian Mathematics and Astronomy. Aryabhatya worked on the place value system using to signify numbers and stating.

His famous work are the" Aryabhatiya "and the"Aryasiddhanta".The Mathematical part of the Aryabhatiya covers arithmetic. Algebra, plane and spherical trigonometry. The Arya-siddhanta, a lot work on astronomical computation.

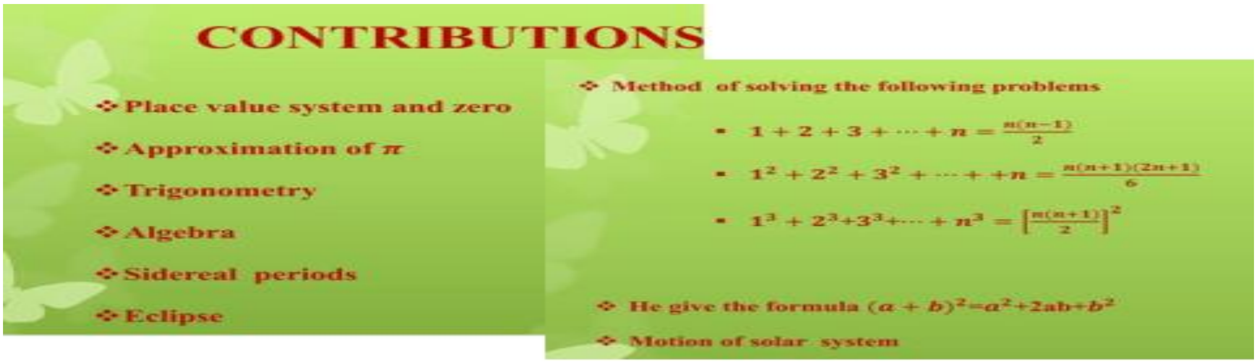
Approximation of Pi: Aryabhata work on the approximation for pi (π) and may have come to the conclusion that π is an irrational number.In the 2nd part of Aryabhatiya; he writes the ratio of circumference to diameter is 3.1416

Aryabhata given the formula for area of a triangle .He also discussed the concept ofsine in his work by the name of ardhajya. If we use Aryabhata's table & calculate the
value of sin300 which is 1719/3438=0.5.,the value is correct. His alphabetic code is commonly known as the Aryabhata cipher.

He was first person to say that Earth is spherical and it revolves around the sun.

He gave the formula (a+b)2 =a2 + b2 + 2ab.

- He taught the method of solving the following problems:
- $1+2+3+\ldots\ldots\ldots\ldots+n = n(n+1)/2$
- $1^2 + 2^2 + 3^2 + \ldots\ldots\ldots + n^2 = n(n+1)(2n+1)/6$
- $1^3 + 2^3 + 3^3 + \ldots\ldots\ldots\ldots\ldots + n^3 = (n(n+1)/2)^2$



BRAHMAGUPTA (598-668 AD)

The most significant contribution of Brahmagupta was the introduction of zero(0) to the mathematics which stood for nothing.Brahmagupta was born in 598 A.D.in Bhinmal city in the state of Rajasthan. He was a mathematician and astronomer, who wrote many important works on mathematics and astronomy. His best known work is the “Brahmasphuta-siddhanta”, written in 628 AD in Bhinmal.

He was the first to use zero as a number. He gave rules to compute with zero.

He gave four methods of multiplication.

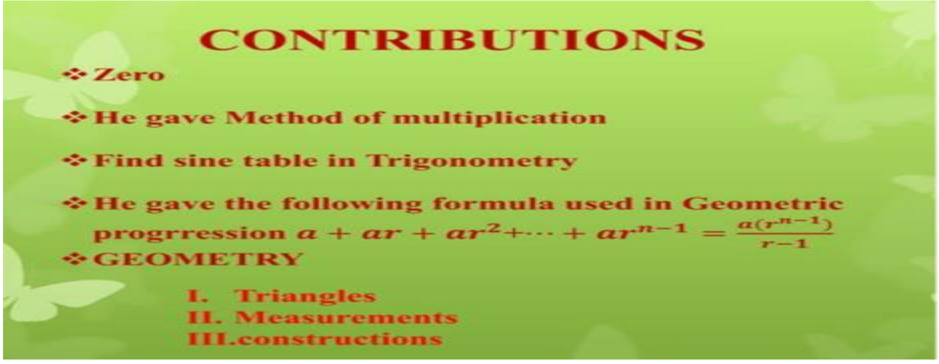
He gave following formulae, used in G.P.

series $a+ar+ar^2+ar^3+.........+ar^{n-1}=a(r^n -^1)/(r-1)$.

He gave the following formulae(Brahmagupta’s formula):

Area of a cyclic quadrilateral with side a,b,c,d =9(s-a)(s-b)(s-c)(s-d),

where 2s= a+b+c+d.



BHASKARACHARYA (1114-1185 AD)

He was born in Bijapur in modern Karnataka. He & his work represent a significant contribution to mathematical & astronomical knowledge in the 12th century.His main work “Siddhanta Shiromani” is divided into four parts called Lilawati, Bijaganit, Grahaganita and Goladhyaya. These four sections deal with arithmetic, algebra, mathematics of planets and spheres respectively.

He was the first to give that any number divided by zero gives infinity.

He was written a lot about zero, surds, permutation and combination.

He wrote,” The hundredth part of the circumference of a circle seems to be straight. Our earth is a big sphere and that’s why it appears to be flat.”

He gave the formulae like: sin(A±B)=sinAcosB ± cosAsinB.

He calculated derivatives of Trigonometric functions and formulae.

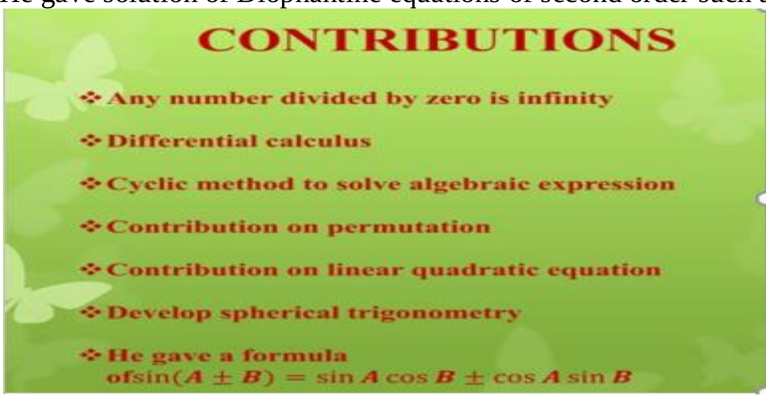
He developed spherical trigonometry along with a number of other trigonometric results.

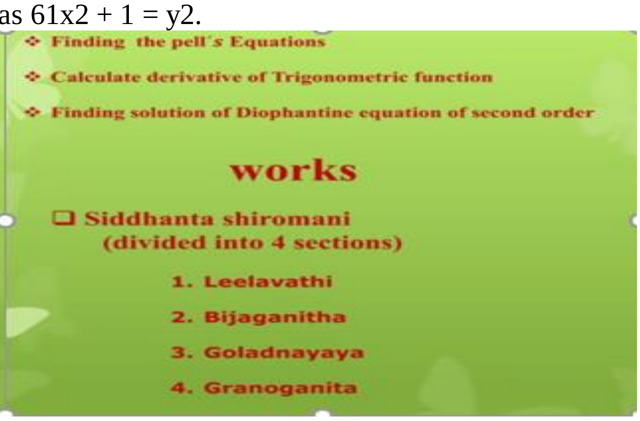
He explained solution of quadratic, cubic and quartic indeterminate equations.

He developed a proof of Pythagoras Theorem by calculating the same area in two different ways & these cancel out terms to get a2 + b2 = c2.

He gave first general method for finding the solution of the problem x2 – ny2 = 1 (so called Pell’s equation).

He gave solution of Diophantine equations of second order such as 61x2 + 1 = y2.





Bhaskara I

He was born in the 7th century CE in Maharashtra’s Parbhani district.

- He was the first mathematician to depict zero with a circle.
- He gave a notable rational approximation of the sine function in his work Aryabhatiyabhashya, a commentary on Aryabhata’s work.

- His other works are Mahabhaskariya and Laghubhaskariya.

Mahavira

- He was born in Gulbarga, Karnataka.
- His work Ganita Sara Sangraha contains more than 1,130 rules and examples separated into nine chapters on basic operations, reductions of fractions and linear or quadratic equations having one unknown.
- He developed a formula for calculating the area and perimeter of an ellipse.
- He has also found methods to estimate a number's square and cube roots.

Bhaskara II

- He was born in Bijapur, Karnataka in 1114 CE.
- He is celebrated as the greatest mathematician of the medieval period.
- He devised differential coefficient and differential calculus, prior to Newton and Leibniz.
- His famous work Siddhanta Shiromani has laid the base for various theories in arithmetic, algebra, etc.
- He has also written a treatise named Karana Kautuhala.

RAMANUJAN (1887-1920)

Ramanujan was born on 22nd of December 1887 in Erode, Madras Presidency. He made extraordinary contributions to mathematical analysis, number theory, infinite series, and continued fractions.He demonstrated unusual mathematical skill at school,winningaccolades and awards.

By 17,he had conducted his own mathematical research on Bernoullinumbers and the Euler-Mascheroni constant.

He discovered theorems of his own and rediscovered Euler's identity independently.

He independently compiled nearly 3900 results (mostly identities andequations).Nearly all his claims have now been proved correct.

Ramanujan Showed that any big number can be written as sum of not more than fourprime numbers.

He showed that how to divide the number into two or more squares or cubes.

Ramanujan'sNumber:WhenMr.G.H. Hardy came to see Ramanujan in taxi number729,Ramanujan said that 1729 is the smallest number which can be written in theform of sum of cubes of two numbers in two ways,i.e.1729=93+103= since thanthe number 1729 is called Ramanujan's number.

In 1918, Ramanujan and Hardy studied the partition function P(n) extensively andgave a non-convergent asymptotic series that permits exact computation of thenumber of partition of an integer.

He discovered mock theta function in the last year of his life .For many years thesefunctions were a mystry,but they are now known to be the holomorphic parts ofharmonic weak mass forms.

C.R.Rao

- C.R. Rao is an Indian-American statistician and professor emeritus at Penn State University.
- He is known for pioneering work in multivariate analysis, statistical genetics, and DNA sequencing.
- He developed the Rao–Blackwell theorem and the Cramér–Rao bound.
- He was awarded the National Medal of Science in 2010. It is the highest scientific honor in the U.S.
- He was the president of the International Statistical Institute from 1985 to 1987.
- He was awarded the Padma Vibhushan in 2013 for his contributions to science.

SatyendraNath Bose

- He was born in Kolkata in the year 1894.
- He was given the prestigious Padma Vibhushan award.
- He was a theoretical physicist, laying the foundation for theoretical physics in India.
- He was known for his work on quantum mechanics, providing a base for Bose-Einstein statistics.
- Some of the other contributions of SatyendraNath Bose include Bose-Einstein condensate, Boson, Single field theory , Diffraction of X-ray and research in the field of Electromagnetic waves.

Shakuntala Devi

- She was born in 1929 in Karnataka's Bengaluru.

- She was a mental calculator commonly known as the “Human Computer”.
- She competed against the world’s then fastest computer, UNIVAC to calculate the 23rd root of a number with 201 digits and won by 10 seconds.
- She earned a place in The Guinness Book of World Records, 1982 edition.

D.R.Kaprekar

- DattatreyaRamchandraKaprekar was born in Dahanu in 1905.
- He was an Indian recreational mathematician describing multiple classes of natural numbers that included the Kaprekar, Harshad and self numbers.
- He discovered Kaprekar’s constant named after him.
- 

Narendra Karmarkar

- Narendra Karmarkar is an Indian-American mathematician known for Karmarkar's algorithm. It is a polynomial time algorithm for linear programming problems.
- He developed this algorithm in 1984 while working at AT&T Bell Laboratories. It was a revolutionary improvement over the then-existing algorithms.
- Karmarkar's algorithm is exponentially faster than the simplex algorithm for large-scale linear programming problems.
- It laid the foundation for several new interior-point algorithms widely used today.
- He received the John von Neumann Theory Prize in 1989 for developing Karmarkar's algorithm.
- He retired from AT&T Bell Laboratories as the director of theoretical research in 1998.

ASTRONOMY

ASTRONOMY = Khagolashastra ;-Astronomy, science that encompasses the study of all extraterrestrial objects and phenomena.Astronomy, science that encompasses the study of all extraterrestrial objects and phenomena

The study of the planetray motion and relative positions of celestial body.

A celestial body is a natural object outside of the Earth’s atmosphere. This can include planets, sun ,stars, moons, asteroids, comets, **Satellites:** , or even nebulae and galaxies.

Astronomy is the study of space, celestial objects, and everything that isn't on Earth. This includes the study of the movement of stars, where the Earth lies in relation to other planets and celestial objects, and what makes up stars and other planets

Ancient astronomers' knowledge about planets wasn't as in-depth as our current knowledge, but they were able to differentiate many of the planets from the stars. In particular, they knew about the five planets that were visible to the naked eye: Mars, Venus, Saturn, Jupiter, and Mercury.

Indians learnt many elements of astronomy from the Greeks and the Romans due to closer contacts of India with Greco-Roman world during the post-Mauryan age. Astronomy was the only branch of knowledge in which Indians learnt from foreigners.

Ancient Indian astronomy may be classified into two main
categories:

(1) the vedic astronomy and
(2) the post vedic astronomy.

The vedic astronomy is the astronomy of the_vedic period i.e. the astronomy found in the vedicsamhitas and brahmanas and alliedliterature

Post-Vedic astronomy in ancient India focused on the sun and moon, and the study of natural divisions of time caused by their motion. These divisions included days, months, seasons, years, new moons, full moons, equinoxes, and solstices.

Post-Vedic astronomy is found in literature such as the Upanishads, Brahmanas, Smritis, and Puranas, which may have evolved over centuries with some additions as late as 2,300 years ago. These texts include stories and references from the post-Vedic era

The first mention of astronomy in India was in the Vedas, the oldest scriptures of Hinduism, way back in 2nd MilleniumBC(2000 BC to 1001 BC)! Back then, Astronomy was known as “Khagola-Shastra” or the Science of Cosmos (Khagola meaning Cosmos and Shastra meaning Science). Although the oldest Veda manuscripts are dated 1500BC

The earliest reference of Indian interest in the field of astronomy has been found in Taittiriya Brahmana. It makes a mention of Sun, Moon, Nakshatras and seasons.

The JyotisaVedanga, the first Vedic text to mention astronomical data, records events as far back as 4000 BCE, though many archaeo astronomers believe it may include observations as far back as 11 000 BCE.

There is a reference of rules of making a panchanga (almanac) in VedangaJyotisha.

Vedic people knew Jupiter and Venus. In Mahabharata Mercury, Mars and Saturn are mentioned.

Indian astronomy was influenced by Greek astronomy beginning in the 4th century BCE and through the early centuries of the Common Era, for example by the Yavanajataka and the RomakaSiddhanta, a Sanskrit translation of a Greek text disseminated from the 2nd century

Indian astronomy flowered in the 5th-6th century, with Aryabhata, whose Aryabhatiya represented the pinnacle of astronomical knowledge at the time. Later the Indian astronomy significantly influenced Muslim astronomy, Chinese astronomy, European astronomy, and others. Other astronomers of the classical era who further elaborated on Aryabhata's work include Brahmagupta, Varahamihira and Lalla. An identifiable native Indian astronomical tradition remained active throughout the medieval period and into the 16th or 17th century, especially within the Kerala school of astronomy and mathematics

GRAVITY WAS DISCOVERED BY INDIANS NOT NEWTON.

Following is summary of Indian ASTRONOMY-

1. The calculation of occurrences of eclipses
2. Calculation of Earth’s circumference
3. Theorizing about gravity
4. Determining that Sun is a star
5. Determining the number of planets in the Solar System

Use of Astronomy;

1. **Tracking seasons, so that people better understood when the best times were to plant crops leading to more efficient planting and harvesting**
2. **Direction determination**
3. **creation of maps and allowed people to know where they were going and how to get back home,**
4. create astrological charts and read omens(a sign of something that will happen in the future)
5. Developing sophisticated mathematical models and many intriguing theories,
6. **Calendar**
7. **Determining time/ time keeping**
8. Foretelling the future/forecasting of horoscopes.
9. **Trade between distant nations to take place.**
10. **Religious functions**

INDIAN ASTRONOMERS

S.NO.	ASTRONOMER	YEAR
1	Aryabhata	476–550 CE
2	Varāhamihira	505 CE
3	Brahmagupta	598–668 CE
4	Bhāskara I	629 CE
5	Lalla	8th century CE
6	Śrīpati	1045 CE
7	Bhāskara II	1114 CE
8	Madhava of Sangamagrama	1350-1425 CE
9	Mahendra Suri	14th century CE
10	Nilakanthan Somayaji	1444–1544 CE
11	Achyuta Pisārati	1550–1621 CE
12	Amil Kumar Das	1902-1961
13	Subrahmanyan Chandrasekhar	1910-1995
14	Vainu Bappu	1927-1982

IMPORTANT ASTRONOMERS

Aryabhatta (476-550 CE)

1. Came up with celestial calculations.
2. Calculated earth rotation per solar orbit, days per solar obit, and days per lunar orbit.
3. Very Accurate information without early century sources.
4. Also calculated value of pi and solar year.

• Brahmagupta (598 – 668 CE)

1. Wrote astronomical text, Brahmasphutasiddhanta(628 CE)
2. Consisted of 25 Chapter; explanation of astronomical and mathematics concepts.
3. Covered various topics such as conjunction of the planets and problems of diurnal rotation
4. Observed Earth is spherical and moving.
5. Justified the illumination of the Moon by the Sun
6. Defined zero as the result of subtracting a number from itself
7. Gave correct equation of parallax.

Varahamihira (505 – 587 CE)

• Studied various astronomy like Greek, Egyptian, Roman.
• Recognized the force of object falling to the earth known as Gurutvakarshanin Indian terms.
• Made close approach to the concept of heliocentric.

Bhaskara I (600 – 680 CE)

• Wrote Siddhantasoramani, consisted of Goladhyaya (sphere) and Grahaganita (mathematics of the planets).
• Calculated the time earth takes to orbit around the sun to 9 decimal places.
• Mostly contributed to Indian mathematics defining the first Indian decimal system

VARAHAMIHIRA (~500CE)

- **Varahamihira (505 AD – 587 AD) was also a great astronomer in India.**

1. **described the Earth as spherical and as being suspended in space**
2. **Used latitude and longitude**
3. **Described the stars as "fixed" and gave the planets a constant speed.**

ARYABHATA I (~500CE)

- Real development in the field of astronomy began in around 500 AD. The first person to make significant contribution to it was Aryabhata (476 AD – 550 AD) who wrote a book titled Aryabhatiyam.
- The Indian astronomer-mathematician Aryabhata (476–550), in his magnum opus Aryabhatiya, propounded a mathematical heliocentric model in which the Earth was taken to be spinning on its axis and the periods of the planets were given with respect to a stationary Sun.
- He was also the first to discover that the light from the Moon and the planets were reflected from the Sun, and that the planets follow an elliptical orbit around the Sun, and thus propunded an eccentric elliptical model of the planets, on which he accurately calculated many astronomical constants, such as the times of the solar and lunar eclipses, and the instantaneous motion of the Moon (expressed as a differential equation)
- Aryabhata estimated the circumference of earth and his estimation was very close to modern calculations.
- He postulated a new theory that the earth was round and that it was rotating on its own axis.
- Aryabhata was also the first to explain the true causes of solar and lunar eclipses. He said eclipses were caused by relative motion of sun, moon and earth and that shadows of moon and earth were responsible for eclipses.
- Aryabhata also put rules of planetary movement.
- Heliocentric (somewhat) idea ;- referred to or measured from the sun's center or appearing as if seen from it.
- 1000 years before Copernicus and Galileo
- Described methods of eclipse prediction.
- Knew of most of the planets (at least 7) without a telescope

BRAHMAGUPTA (~600CE)

Brahmagupta (598 AD – 668 AD) was another great astronomer in India.

He wrote Brahmasphutasiddhanta and KhandakhadyakHe calculated the instantaneous motion of a planet, gave correct equations for parallax, and some information related to the computation of eclipses.

He also theorized that all bodies with mass are attracted to the earth. (Around 665 CE, approximately 2351 years before Newton proposed the Law of Gravitation in 1687 AD!)

He emphasized that it was the nature of earth to attract things towards itself. This theory was revolutionary in nature.

He was a precursor to Newton by proclaiming that all things fall to earth automatically. He said that it was a law of nature.

Estimated the Earth's circumference as 5000 yojanas (yojana = 4.5 miles 7.24205 kilometers) Surprisingly accurate! (off by less than 10%)

<u>DEVACHARYA (~600CE)</u>

First astronomer to describe a method for calculating the precession of the equinoxes(An equinox is an event in which a planet's subsolar point passes through its Equator. The equinoxes are the only time when both the Northern and Southern Hemispheres experience roughly equal amounts of daytime and nighttime.)

Bhaskara (1114–1185) expanded on Aryabhata's heliocentric model in his treatise Siddhanta-Shiromani, where he mentioned the law of gravity, discovered that the planets don't orbit the Sun at a uniform velocity, and accurately calculated many astronomical constants based on this model, such as the solar and lunar eclipses, and the velocities and instantaneous motions of the planets

<u>BHASKARA II</u>

- He composed the Siddhantasiromani, which is divided into two parts: Goladhyaya (sphere) and Grahaganita (mathematics of the planets).
- He also calculated to 9 decimal places the time it takes the Earth to orbit the Sun. At the time, the Buddhist University of Nalanda offered formal courses in

<u>INDIAN CALENDAR</u>

A calendar is a chart, device or piece of software which displays the date and the day of the week, and often the whole of a particular year divided up into months, weeks, and days.

Various types of Calendar follow across the World. All types of Calendar system worldwide can be categorized, based on astronomical years which depends on movement of celestial bodies, mainly under 3 types as below.

1. Solar calendars
2. Lunar Calendar system
3. Luni-Solar Calendar system

Various Calendars in India

In India, basically, the four types of calendars are followed, which are as follows:

1. Vikram Samvat, also known as the Hindu lunar calendar.
2. Saka Samvat, also known as the Hindu Solar calendar.
3. Hijri Calendar, also known as the Islamic lunar calendar.
4. Gregorian Calendar, also known as the Scientific solar calenda

SOLAR CALENDARS

- Solar Calendar is finalized on the basis of movement of the Earth around the Sun.
- Solar System ignores Moon completely and it is based on Sun only
- One Year in solar system is the time that earth requires to complete one orbit around the sun, which exactly is 365 days 5 hours 48 minutes and 46 seconds i.e 365.242199 days.
- It has 365 days in a year but every fourth year is a leap year i.e. an extra day is added to the month of February.
- Julian Calendar, Gregorian calendar, Malayalam calendar and Thai calendar are some of the examples of solar calendar.
- Indian Solar Calendars used in the areas of Assam, Bengal, Haryana, Kerala, Punjab, Orissa, Tamil Nadu and Tripura. An actual solar calendar doesn't consists of number of days, hence in order to synchronize, days are summed up and leap years are formed. Indian National Calendar comes under this category.

Lunar Calendar system

- Lunar system ignores Sun completely and it is based on Moon only.
- It is based on Movement of the moon around the earth
- The lunar calendar uses the phases of the moon to measure time, usually measuring the time from new moon to new moon as one month.
- One revolution of the moon around the earth equals one month.
- It has 12 lunation or months comprising of 29 to 30 days , 354 days in total

- In all the regions, each month is divided into two halves
- The dark half of the month is known as Krishnapaksha
- The Krishnapaksha begins with the new moon and ends with a full moon
- The bright half of the month is known as Shuklapaksha
- The Shuklapaksha begins with the New Moon and ends with the full moon
- In a cycle of five years, every third and fifth year has thirteen months

Luni-Solar Calendar system

- Luni Solar system takes both Sun and Moon into consideration.
- By definition, a lunisolar calendar is based on the orbital motion of the Sun and the Moon as observed from the Earth (i.e. in a geocentric model).
- It uses the solar year but divides it into 12 lunar months.
- The definition of a lunar month is the time required for the moon to orbit once around the earth and pass through its complete cycle of phases.
- At the same time this month also considers successive entrances of the sun into 12 rashis – signs of the zodiac, i.e. the 12 constellations marking the path of the sun.
- Indian Calendars used in Andhra Pradesh, Bihar, Gujarat, Karnataka, Kashmir, Madhya Pradesh, Maharashtra, Rajasthan, Uttar Pradesh

PANCHANG

The Hindu calendar year is based on the Vikram era, after King Vikram of Ujjain. The system is still widely used in Northern and Western India. The calendar began in 57-56 BCE.

The Hindu calendar system is based upon the motion of the moon. Each lunar year comprises twelve months. The lunar year comprises 354 days, compared to 365 ¼ days of the Gregorian calendar, which is based on the solar system

In the Indian calendar, seasons follow the sun; months follow the moon; and days, both the sun and the moon.

Two systems of month-reckoning were prevalent in different parts of India at different times:

the purnimanta system - in which the month ends with a full moon; and the

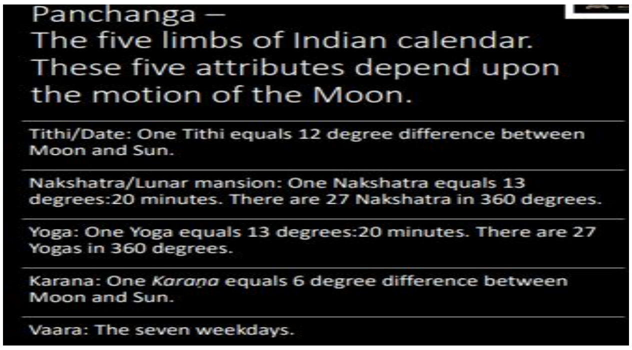
amasanta system - in which the month ends on the darkest night of the month, a 'no moon' night. At present, Gujarat and most of India adhere to the amasanta system, i.e. in which a new month begins with a new moon.

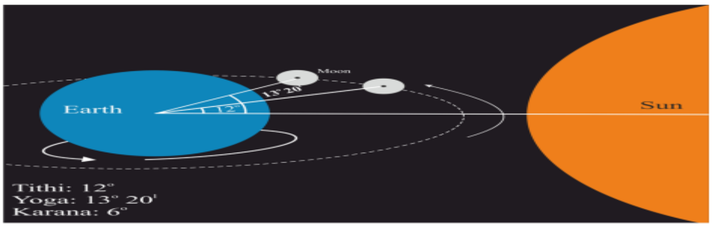
Panchang is a traditional Hindu calendar that has several elements like tithi, nakshatra, yoga, karan, paksha, vaar, etc. These elements are used to determine auspicious timings for various Hindu rituals, ceremonies, and festivals.

The Panchang calendar is derived from two words - Panch means five and ang mean aspects. Together the Calendar means that it uses five aspects to function, i.e.

Panchang = Panch (Five)+ Ang (Limbs)

1. Week or Vara
2. Yoga
3. Karana
4. Lunar days or Tithi(date)
5. Constellation or Nakshatra





Tithi: is the time taken by the moon in increasing its distance from the sun by 12 degrees.
Yoga: The period of time during which the distance between the Sun and the Moon is increased by 13o 20' [13 degrees and 20 minutes], which is about one Day. There are 27 Yoga in 360o .
Karana: Karna is the half of tithi, i.e. Tithi is 12° increase is in a distance between sun and moon and Karana is 6° increase in distance between sun and moon.

Vaaram/Vaar (Days)

The counting of days begins with the sunrise of the first day to the sunrise of the second day. In astrology, this system is known as Ahoratra, which means day and night taken together. Each day consists of 24 Horas. Vāsara refers to the weekdays and the names of the week

Thithi & Karana (Lunar days)

A tithi is a lunar day. There are 15 tithis in the waxing cycle of the moon (shukla paksha), and there are 15 tithis in the waning cycle of the Moon (Krishna paksha). The tithi is based on a relationship between the Sun and the Moon. The first tithi is 12 degrees of the Moon away from the Sun after the new Moon (Amavasya) or full moon (Purnima). The second tithi is the next 12 degrees of the Moon away from the Sun, 12 to 24 degrees. A particular day is ruled by the tithi at sunrise, but the tithi can change any time of the day or night as it is not based on a solar day.

A Karana is half a thithi, or each thithi is divided into two equal parts, each being a Karana known by a specific name. There are 11 types of Karanas.

Yoga (Auspicious & inauspicious periods)

Yoga is yet another term present in Panchang. The number of yogas completed at any time is calculated as the sum of the Nirayana (longitudes) of the Sun and the moon.

Yoga = (Longitude of Sun + Longitude of Moon) / 13°20′

There are 27 Yogas as aligned to the Nakshatras but not fixed with any of them.

Days of the Week	सप्ताह के दिन	HindiTeacherOnline.Com
Monday	Somvaar	सोमवार
Tuesday	Mangalvaar	मंगलवार
Wednesday	Budhvaar	बुधवार
Thursday	Guruvaar	गुरूवार
Friday	Shukravaar	शुक्रवार
Saturday	Shanivaar	शनिवार
Sunday	Ravivaar	रविवार

in months since Indian Calendar dates keep changing) are:

1. *Vaishakh* : Apr-May.
2. *Jyeshtha* : May-June.
3. *Ashaadh* : June-July.
4. *Shravan* : July-Aug.
5. *Bhadrapad* :Aug-Sep.
6. *Ashvin* : Sep-Oct.
7. *Kartik* : Oct-Nov.
8. *Margashirsh/Agrahayan* : Nov-Dec.
9. *Magha* : Dec-Jan.
10. *Pausha* : Jan-Feb.
11. *Phalgun* : Feb-Mar.
12. *Chaitra* : Mar-Apr.

a Panchang month has fifteen Tithis or dates in each Paksha.

1. Prathama.
2. Dwitiya.
3. Tritiya.
4. Chaturthi.
5. Panchami.
6. Shashthi.
7. Saptami.
8. Ashtami.
9. Navami.
10. Dashami.
11. Ekadashi.
12. Dwadashi.
13. Trayodashi.
14. Chaturdashi.
15. Poornima or Amavasya depending on the Paksha.

The lunar month is exactly 29 days 12 hours 44 minutes and 3 seconds long and twelve such months create a lunar year of 354 days 8 hours 48 minutes and 36 seconds

For calculation of length of a month, Indian calendar system uses the interval between two consecutive new moons (no moon) or full moons as a basis.

According to Panchang (which is sort of Hindu Almanac) every month has 30 tithis which are equally divided between 2 pakshas.

Sun Calendar
1. Mesha = Aries
2. Vrushabha = Taurus
3. Mithuna = Gemini
4. Karkataka = Cancer
5. Simha = Leo
6. Kanya = Virgo
7. Tula = Libra
8. Vrushika = Scorpio
9. Danasu = Sagittarius
10. Makara = Capricorn
11. Khumba = Aquarius
12. Meena = Pisces

Sun based Calendar = 365 days

Moon based Calendar = 354 days Is further divided Into 15 day period Krnsa paksha (full to new moon) Shukla paksha (new to full moon)

Moon Calendar Mont
1. Chitra
2. Vishaka
3. Jeeshta
4. Ashada
5. Shravana
6. Bhadrapada
7. Ashvayuja
8. Karthika
9. Margashira
10. Pushya
11. Maha
12. Palguna

USE OF CALENDAR

used to determine

1. Time
2. Day
3. Year
4. seasons
5. weather
6. Dates of festivals,
7. Religious ceremonies,
8. Muhurat timing
9. The zodiac signs
10. Grow crops
11. Harvesting time

PUBLIC ADMINISTRATION AND GOVERNANCE;

The Vedic period or Vedic age (c. 1500 – c. 500 BCE) gets its name from the Vedas. . Early Vedic Aryans were organized into tribes rather than kingdoms. The chief of a tribe was called ‘Rajan.’ The main responsibility of the Rajan was to protect the tribe. He was aided by several functionaries, including the purohita (chaplain), the senani (army chief), dutas(envoys), and spash (spies)

The Rig Veda period's administrative units were kull, gramme, vesh, and nation.The Vedic king's rule was aided by the samiti and sabha, two popular bodies. Whenpeople gather in a samiti, it is called an assembly. According to tradition, the king wasrequired to attend the samiti. The sabha also served as the country's court of justice.According to the Vedas, national life and activities were first expressed through publicassemblies and institutions. The king's primary duty was to safeguard his subjects. Underthis administration, Gramini was in charge of the village and Senani was in charge of thearmy. The purohit was the most important of the many ministers who served under theking

The Post- Vedic Period: There were many kingdoms and Mahajanpadas during the Buddha's time.Magadha, Avanti, Vats, and Kaushal were the four major kingdoms, in addition to there publics

The earliest evidence of ancient Indian administration can be found in the Indus Valley Civilization. The administrative system in India has a rich history that dates back to the Indus Valley Civilization, which existed over 5000 years ago. During this time, the king held immense power and was responsible for overseeing all aspects of the kingdom. He was supported by a council of ministers, officers, and other functionaries, which meant that the administration was heavily centralized around the institution of the king.
The administrative systems became more structured and organized with the rise of the Mauryan and Gupta dynasties. Both of these dynasties implemented sophisticated governmental machinery that facilitated the efficient execution of state functions
So the structure was King & Kingdom – There is no mention of Democracy
State Administration includes -
(a) Land Records
(b) Law and Order including classification of crime, courts & punishment
(c) Salary for Employees,
(d) Hospitals and Medical Support for Citizens
(e) Water management,

(f) Transport – Horse, Bullock Carts, Handyman and palanquin

Governance:

King = Svamin ;
Capital = Durg = Citizens lived (inside & adjoining)
Prime Minister for the Kingdom = Maha Mantri
Cabinet Ministers = Mantri Parishad
Minister = Amatya for every Janapada (Territory)
Finance Minister = Kosha Adyaksh
Defence Minister = Danda Mantri

Law & Order :

Courts were classified into:

Criminal Courts & Civil Courts

There were different level of courts

District Criminal Courts;
State Criminal Courts

These courts had respective officers

Superintendents;
Chancellors,
Judges & Jury
King

Mauryan Administrative System

- The Mauryan Empire marked a significant period in Indian history, as it introduced a new era of political and administrative organization.
- The empire was divided into four provinces, each with its own capital: Tosali in the east. Ujjayain in the west. Suvarnagiri in the south, and Taxila in the north. The central capital was Patliputra.
- The Mauryan administrative system was well-organized and comprehensive. It included various levels of administration, such as central, provincial, local, revenue. judicial, and military. Each of these administrative structures played a crucial role in the governance of the empire.

1. Central Administration
2. Provincial Administration
3. Local Administration
4. Revenue Administration
5. Judicial Administration
6. Military Administration

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central administration. it was directly overseen by the king. This level of administration was responsible for implementing the king's policies and ensuring the smooth functioning of the empire.

- He was responsible for the kingdom's safety and security. and he determined the overall policies to be followed by all officials. The King appointed ministers and other officers of the royal administration and served as the supreme commander of the army and head of the entire military.

The provincial administration was responsible for governing the four provinces and maintaining order within their territories.

- The provincial administration worked on similar lines of the central administration.
- The provincial Governors were responsible for day-to-day conduct of administration of provinces. They were expected to consult on important matters. (the central administration). There were also the district officers. reporters. clerks, who helped in the smooth running of provincial administration.

The local administration managed the affairs of the towns and villages within each province.

- This included overseeing public works projects. maintaining law and order, and collecting taxes.
- The revenue administration was responsible for the collection of taxes and other financial matters.
- This branch ensured that the empires finances were in order and that funds were appropriately allocated.

The district administration was in the charge of 'Rajukas'. whose position and functions are similar to today's district collectors.

- He was assisted by `Yuktas' or subordinate officials. In the urban was, there was a Municipal Board with 30 members.
- There were six committees with five Board members in each to manage the administration of cities.

Village administration was in the hands of `Gramanii and his superior was called 'Gopa: who was incharge of ten to fifteen villages. Census was a regular activity and the village officials were to number the people along with other details such as their castes and occupations.

The judicial administration dealt with legal matters and the enforcement of laws. It ensured that justice was served and that disputes were resolved in a fair and timely manner.

- The King served as the head of the judiciary, he appointed judges to handle regular cases.

The military administration was responsible for the defense and security of the empire.

- This branch was responsible for maintaining a strong and well-equipped army to protect the empires borders and maintain internal stability.
- The King was the supreme commander of the military.
- The Mauryan army was well organized and it was under the control of a 'Senapati:
- six committees with five members in each. These committees were responsible to manage the following wings of the military:
- Navy
- Transport and Supply
- Infantry
- Cavalry
- War Chariots
- War Elephants
- Each of the above wings was under the control of Adhyakshai or Superintendents.

Administrative System during Gupta Period

- The administrative system during the Gupta dynasty was found more or less similar to that of the Mauryan Empire.
- The Empire was classified into administrative divisions like Rajya, Rashtra, Desha. and Mandala.
- This denotes the importance being given to administrative decentralization.
- Central Administration
- During the Gupta Age. the government was a benevolent monarchy, with the king holding the highest authority and possessing extensive powers for the smooth functioning of the empire.

- The king was assisted by a council of ministers. known as the Mantri Parishad. which included the Prime Minister (Mantri Mukhya) and other officials responsible for various areas such as military affairs. law and order. and more.
- These officials were known as Mahasandhi Vigrahaka, Amatya. Mahabaladhikari,and mahadandnayak

-
- **Provincial Administration**
- The Guptas organized a system of provincial and local administration.
- The Empire was divided into divisions called thukth' and each Bhukti was placed under the charge of an 'Uparika:
- The Bhuktis were divided into districts or Vishayas and each Vishaya was under a Vishayapati.
- The Vishayapatis were generally members of royal family. They were assisted in the work by a council of representatives.
- **Local Administration**
- The local administration of the city was managed by a Parishad, with its leader referred to as Nagara Rakshaka. Another officer, called the Purapala Uparika, supervised the Nagara Rakshaka.
- In addition, there was a special officer known as Avasthika. who served as the Superintendent of Dharamsalas (rest houses).
- The village represented the smallest administrative unit. The village head, known as Gramika. was supported by other officials. such as Dutas (messengers), headmen, and Kartri.

Revenue Administration

- • Revenue administration played a crucial role in the functioning of the Gupta Empire, with various officials responsible for its proper management.
- These officials included Viniyuktaka, Rajuka, Uparika. and Dashparadhika, among others.
- Their primary responsibility was to ensure the collection of taxes and other forms of revenue for the empire.

Judicial Administration

- Judicial system was far more developed under the Guptas than in earlier times.
- It was the duty of the King to uphold the law and deal with legal cases with the help of Brahman priests. judges, and ministers.
- King was the highest court of appeal.
- At the lowest level of the judicial system was the village assembly or trade guild. • These were the village councils, which were appointed to settle the disputes between the parties that appeared before them.

Military Administration

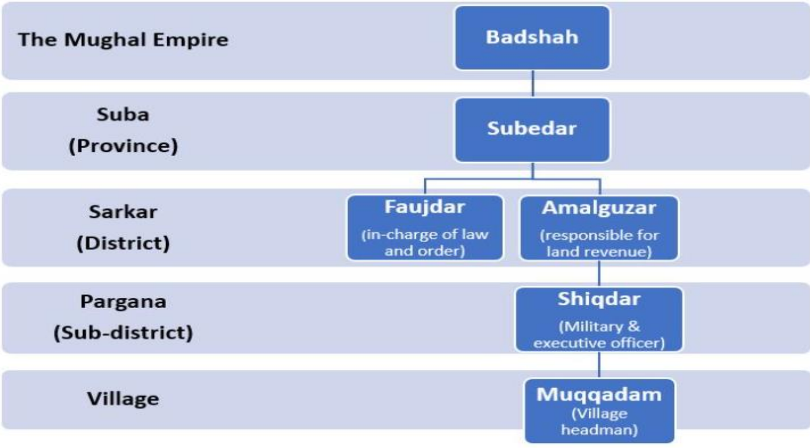
- The Gupta Empire was known for its powerful and expansive military. which included a standing army.
- The empire's territories were closely monitored and guarded by military officers, who held titles such as Senapati, Mahasenapati, Baladhikrita. Mahabaladhikrita, Dandanayaka, Sandhivigrahika. and Mahasandhivigrahika.
- These officers played crucial roles in managing the military administration.
- The Gupta military was organized into four main divisions:

1. the intelligence wing,
2. the cavalry wing,
3. the elephant wing, and
4. the navy.

The Administration of Mughals

- The Mughal empire was famous and kept as a benchmark in the field of administration which they adopted from their ancestors.
- They exhibited a proper structure in administration which was split into three layers.

1. Central Administration.
2. Provincial Administration.
3. Local Administration



- **Central Administration of Mughals:**
- The Emperor was the central authority in Mughal Administration where he holds the supreme power over the law making and Final arbitrator of Justice.
- He also appointed separate officers to look after the affairs of the departments allocated to them.
- The Mughal administration under the emperor had four main divisions.
- Wakil: He is the Prime Minister of the main provinces of the Mughal empire who overlooked the administration of other departments. He was under the direct control of the Emperor.
- Diwan: He is the Chief Minister who worked under the Wakil and the controller of revenue and finance of all the departments functioning in the empire.
- He also assigned the duties and revenue for Faujdars of the Provinc
- **Mir Bakshi**: He was the head of **Military affairs** under the Mughal administration who looked over the recruitment of soldiers, entering their role in the **Mansabdari** system, and maintaining their ranks.
- **Mir Saman**: Also known as the **Khan-i-Samman** look after the royal household. He maintained the records of "**Kharkhanas or royal workshops**".
- **Sadr**: He was the head of **religious departments**, collected charity and other revenue to enhance education in Madras

The Mughal Provincial Administration:

- The Mughal emperors divided their provinces named "Subas" which was headed by "Subedars".
- The Suba's are further divided into three groups.

1. The Subas are further divided into Sakars or districts,
2. and when the group of villages are combined they are called Parganas.
3. Villages or Grama are the lowest unit of administration under Mughals.

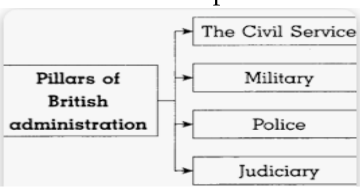
Local Administration:

- The small towns and cities are administered by Kotwals, who are the law officers of the districts and Faujdar was the head of the Sarkar.

The Village heads are known as "Muquaddams" who dealt with the local administration of villages

BRITISH ADMINISTRATION

The British Raj was the period of British rule over the Indian subcontinent from 1858 to 1947,. The British Raj was based on three pillars: the Civil Service, the Army, and the Police:



Civil services:

- The **Civil servants** are officials who implemented the laws enacted by the British company, and aggregating the revenue were their primary duty. he British introduced the name 'Civil servant' in 1829 to distinguish the civilian heads from the military personnel under the company.

Army of the British:

The army played a stellar role in the triumph of the British company and its dominance in India. he army served as the second crucial tenet of the British government in India. It served as the primary tool for extending and defending the British Empire in Asia and Africa. It was the means by which the Indian powers were subdued. It protected the British Empire in India from foreign adversaries

It waged several wars under the three presidencies.

1. Army of **Bengal** Presidency
2. Army of **Madras** Presidency and
3. Army of **Bombay** Presidency

. Police

4. The police, another institution founded by Cornwallis, served as the third pillar of British power.
5. He established a permanent police force to uphold peace and order and removed the zamindars of their policing duties

Judicial Organization

6. Through the establishment of a hierarchy of civil and criminal courts, the British set the groundwork for a new system of delivering justice.
7. Though given a start by Warren Hastings, the system was stabilized by Cornwallis in 1793. In each district was established a Diwani Adalat, or civil court, presided over by the District Judge who belonged to the Civil Service

Governance & Administration present

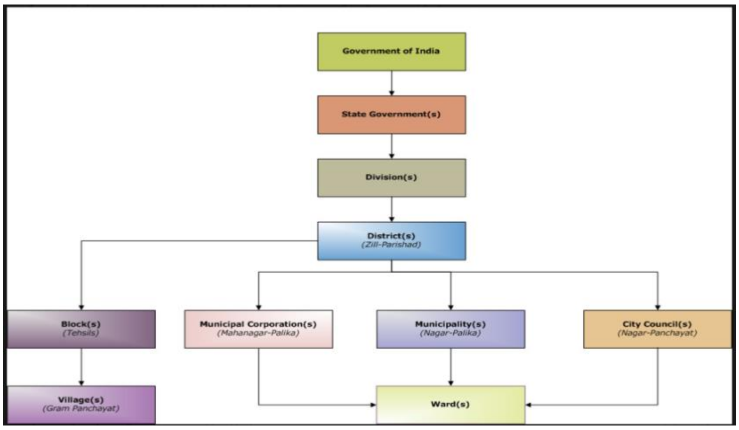
India is a Sovereign Socialist Secular Democratic Republic with a Parliamentary form of government which is federal in structure with unitary features.

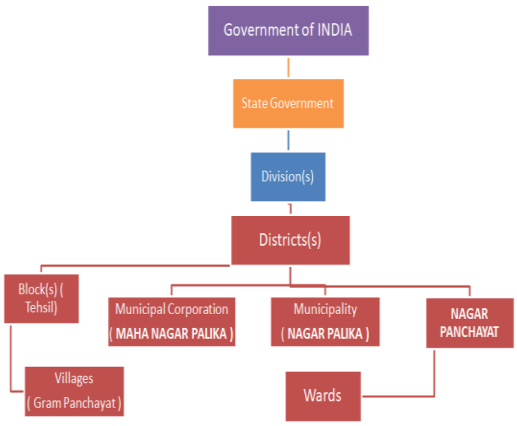
There is a Council of Ministers with the Prime Minster as its head to advice the President who is the constitutional head of the country.

Similarly in states there is a Council of Ministers with the Chief Minister as its head, who advices the Governor.

Indian governance and administration at

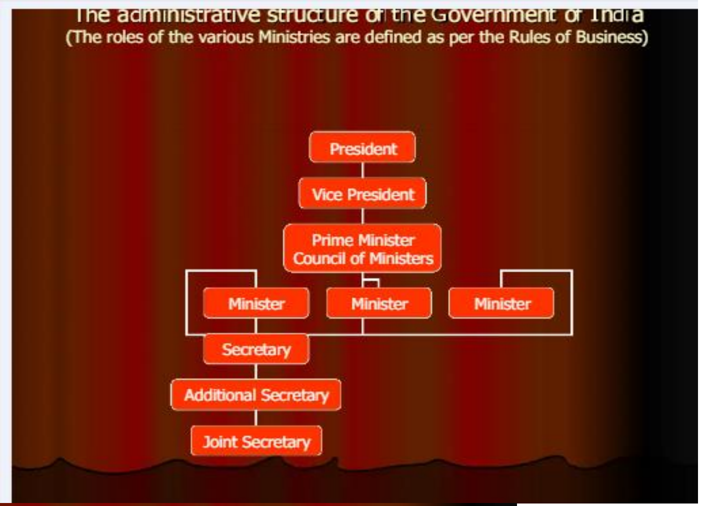
1. the Central,
2. state
3. local level.

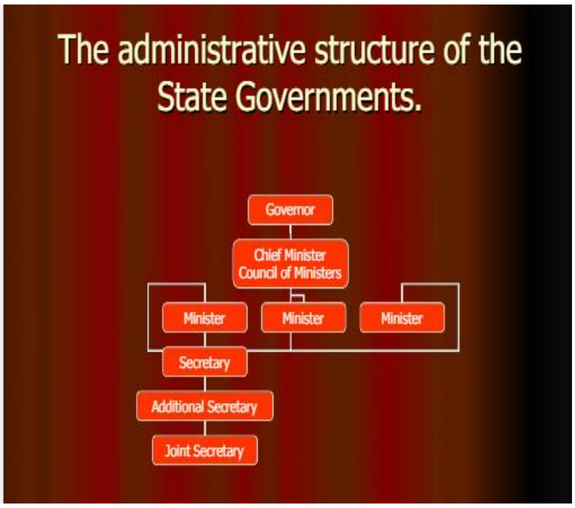


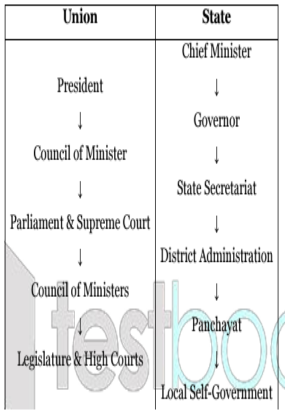


Functions of Indian Administrative Systems

- Indian administration is accountable to the people of India.
- The main function is to act in accordance with objectives laid down in Preamble of the constitution, DPSP, FR.
- To secure all its citizens' liberty, equality and fraternity.
- To work for socio economic welfare of society
- To formulate and implement the projects for the economic development
- Work for government instead for the political party in power







Union	State
	Chief Minister
President	↓
↓	Governor
Council of Minister	↓
↓	State Secretariat
Parliament & Supreme Court	↓
↓	District Administration
Council of Ministers	↓
↓	Panchayat
Legislature & High Courts	↓
	Local Self-Government

TRADE,

- The buying or selling of goods or services between people or countries
- India has thrived in trade since 2600 BC. During Harappan civilization,In ancient times, trade began as a barter system in which people exchanged one object for another. Portugal was the first country to establish trade relations with India
- During ancient times, Indiana's were the masters of the seaborne trade of Europe, Asia and Africa.
- Commercial cities like Harappa and Mohenjodaro were some examples for the business development of ancient India
- These civilizations had established commercial connections with Mesopotamia and traded in gold, silver, copper, gemstones, beads, pearls, sea shells etc.
- Large extent its supplies of fine cotton and silk fabric, spices, indigo, sugar, drugs, precious stones and many curious works of art from India in exchange of gold and silver
- Trade and commerce have played a vital role in making India to evolve as a major player in the economic world in ancient times. Archaeological evidences have shown that **trade and commerce was the basis of the economy of ancient India**
- The progress of domestic and international trade in ancient India developed due to flow of wealth through these routes.
- Many trade centres and Industrial Belt also started. Archaeological evidences have shown that trade and commerce was the base of the economy of ancient India.
- The trade was mainly carried out by water and land.

TRADING COMMUNITIES

Ancient India had many trading communities, including guilds, merchants, and other groups:

Gujarati traders: Hindu Baniyas and Muslim Bohras traded with the Red Sea, Persian Gulf, East Africa, Southeast Asia, and China.

Punjabi and Multani merchants: These merchants handled business in the northern region.

Bhats: These merchants managed trade in Gujarat and Rajasthan.
Mahajan: This group was found in western India and was represented by their chief, the nagarseth, in urban centers like Ahmedabad.
Chettiars: This community was an important trading group in South India.
Manigramam and Nanadesi: These guilds in South India traded within the peninsula and with China and South West Asia.
Marwari Oswal: This community became a principal trading group of the country.
Wholesale traders: These traders were known as "seth" or "bohra".
Retail traders: These traders were known as "beoparis" or "banik".
Banjaras: This special class specialized in trading.

MAJOR TRADE CENTRES IN ANCIENT INDIA

Patliputra: Known as Patna today. It was not only a commercial town, but also a major centre for export of stones.
Peshawar: It was an important export centre for wool and for import of horse. It had huge commercial transactions between India, China and Rome.
Taxila: It was a major centre of land route between India and Central Asia. It was also a city of financial and commercial banks. It was a place of Buddhist centre of learning. The famous Taxila university flourished here.
Indraprastha: It was a commercial junction on Royal Road. Where most routes leading to east, west, south and north converged.
Mathura: It was an emporium of trade. Many routes from South India touched Mathura and Bharuch.
Varanasi: t grew as major centre of textile industry and became famous for beautiful gold silk cloth and sandalwood workmanship. It had link with Taxila and Bharuch.
Mithila: Mithila established trading colonies in South China especially in Yunnan.
Ujjain: Agate, Carnelian, muslin and mallow cloth were exported from Ujain to different centres.
Surat: Textile of Surat were famous for their gold border (zari). Surat's hundis were honoured in far off markets of Egypt and Iran.
Kanchi: Here Chinese used to come in foreign ships to purchase pearls, glass and raw stones and in return they sold gold and silk.
Madura: It was the capital of Pandayas. It attracted foreign merchants particularly Romans for carrying out overseas trade.
Kaveripatta: also known as Kaveripatnam, it was scientific in its construction as city provided loading, unloading and strong facilities of merchandise. It was a centre of trade for perfumes, cosmetic, scents, silk, wool, cotton, corals, pearls, gold and precious stones.
Tamralipti: It was one of the greatest port connected both by sea and land with west and far east. It was linked by road to Banaras and Taxila.

TRADE ROUTE

There were two types of trade rout

1. Internal
2. External

Internal

The two major trade routes of Indian subcontinent were Uttarpath of the north and north-west and Dakshinpath of the central India and the southern peninsula

1. UTTARPATH
2. AND DAKSHINPATH

UTTARPATH is also referred as "Northern High Road". This trade route included the whole of **North Western India, Northern India, Eastern India and also Himalaya in the north to Vindhya in the south**.
The route connected Bengal, Bihar, Gorakhpur, Bijnor, Saharanpur, Jallunder, Lahore, Taxila etc. Its Southern sector connected Lahore, Delhi, Hastinapur, Varanasi, Allahabad, Patliputra and Rajgir etc.
DAKSHINPATH, is often referred as "Southern High Road" connected Varanasi, Ujjain, Narmada Valley and Pratisthana (Paithan — Maharashtra) and then to western coast of India, running in southern direction

External

1. silk route
2. The Spice Route was a way of maritime trade.
3. The Salt Route,

4. Incense Route,
5. Tin Route,
6. The Amber Road.
7. Tea route

Silk route

- India, the Silk Route was a series of trade routes that connected the Indian subcontinent with the rest of Asia and Europe.
- These routes were important for the trade of silk, spices, precious stones, and other luxury goods that were highly valued in the ancient world.

This was one of the most famous trade routes of ancient times that connected China to Rome. Through Silk trade route, silk was traded to Rome from China in exchange of gold, silver and wool. It was originated from Xi'an in China, travelled

- Land routes were, of course, part of this network, especially the famed Silk Road through which countless caravans of traders crossing many kingdoms connected the Mediterranean world with Asia.

THE FAMOUS SILK ROUTE

Spice Route in the 15th century

17.3.1 Spice Route

it is a distance of over 15,000 kilometres and, even today, is not an easy journey.

The sea route from west coast of Japan to Indonesian islands to India (around) to Middle East to Mediterranean area (across) to Europe is called Spice Route.

Incense Route

Romans, Greeks and Egyptians were very fond of Frankincense and myrrh which was found in Yemen and Oman (Southern end of Arabian Peninsula). Frankincense and myrrh was burned as incense or used as perfumes. It was also used during burial rituals to aid embalming. The Incense Route was developed to transport valuable incense to the Mediterranean by Arabs through camel. This incense

Amber Road (Trading Beads)

Romans have developed Amber Road for transporting amber beads from the Baltic to the rest of Europe. Amber beads were used as decorative item and valued for medicinal purposes. Baltic Sea is famous for good quality amber beads which were formed millions of years ago when covered forest submerged. Traces

Tea Route (Tea-Horse Road)

Tea route was developed for transporting Chinese tea and Tibetan warhorses from Hengduan Mountains of China through Tibet to India. This route was most dangerous due to terrain of crossing areas and numerous rivers. During Song

Major Exports and Imports

Exports consisted of spices, wheat, sugar, indigo, opium, sesame oil, cotton, parrot, live animals and animal products—hides, skin, furs, horns, tortoise shells, pearls, sapphires, quartz, crystal, lapis, lazuli, granites, turquoise and copper etc.

Imports included horses, animal products, Chinese silk, flax and linen, wine, gold, silver, tin, copper, lead, rubies, coral, glass, amber, etc.

Indigenous Banking System

- The indigenous banking system in ancient India dates back to the Vedic period (1500 BCE–600 BCE) and was a major part of the financial system until the mid-19th century.
- The system was based on guilds, or sreni, which were informal financial institutions that promoted trade within communities.
- Indigenous bankers, or ranas, were individuals or private firms that accepted deposits, gave loans, and dealt in hundis, a popular instrument of exchange.
- Indigenous Banking system • An indigenous banker or bank is defined as an individual or private firm which receive deposits, deals in hundies or engages itself in lending money.
- In ancient India indigenous banking system played a prominent role in lending money and financing domestic and foreign trade with Currency and Hundi.
- As economic life progressed, money served as a medium of exchange.
- The introduction of metallic money and its use accelerated economic activities. With the development of banking, people began to deposit precious metals with lending individuals functioning as Bankers or Seths and borrowed money from them.
-
- Documents such as Hundi and Chitti were in use for carrying out transactions in which money passed from one hand to another hand.

❶Use of Metals as money: metals were used as money due to high durability and divisibility, it served as a medium of exchange

❷Use of Hundi and Chhitti Documents were in use for carrying out transactions in which money passed from hand to hand. Hundi is an unconditional contract or order which warrantees a monetary payment which can be transferred by a valid negotiation

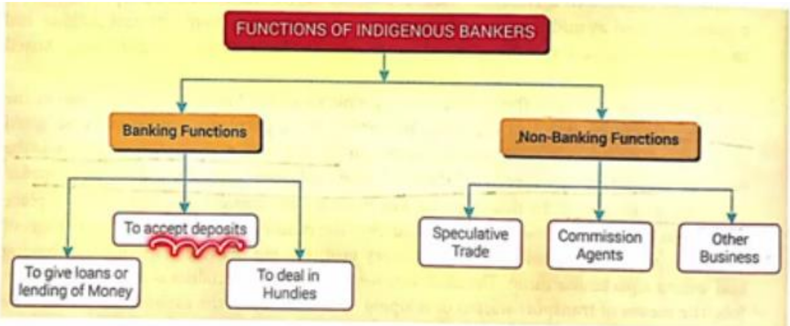
❸Development of Banking:Banks started lending money and financing the domestic and foreign trade and it encouraged people to deposit precious metals with lending individuals.

❹Agriculture and livelihood opportunities:Agriculture and domestication of animals were important sources of livelihood, along with this people also relied on weaving cottonon, dyeing fabrics, making clay pots, etc.

Development of Indigenous banking system played a special role.
(i) People began to deposit precious metals with lending institutions acting as bankers or seths.
ii) These bankers and seths became an instrument for supplying money for producing more goods.
iii) Indigenous banks played a special role in lending money and financing domestic trade.
Agriculture and Domestication of animals also help to generate surplus and savings for investment
Due to favourable climatic conditions people of India were able to raise two or three crops in a year.
In addition to agriculture ; weaving, dyeing fabrics, making clay pots, sculpting, cottage industries, masonry manufacturing, transport (carts, boats, etc.) were able to generate surplus for investment.

Functions of indigenous banks

1. Accepting deposits
2. Advancing loans
3. Discounting Hundis
4. Speculation
5. Financing inland trade



Banking function

I. lending of Money: Money-lending is the principal banking function of the indigenow bankers. Loans are given on the basis of promissory notes, simple receipts, legal documents personal security and mortgages. Rate of interest is different for different persons and a: different places. Rate of interest depends largely on the nature of security offered. Gooa security (in the form of gold, silver, ornaments etc.) keeps the interest rate to about let while on the poor security the rate of interest ranges between 24-40%. Indigenous bankers give both short-period loans to agriculture, trade and industry.

2. Accepting deposits;- Another important banking function of the indigenous bankers is accept deposits of the people. These deposits arc paid on demand or at the stipulated order. Deposits can be withdrawn by the depositors personally or through their written saorders. Some indigenous bankers use cheque book system also, but this system is confined to only few places. For lending money, the indigenous bankers do not depend much on their deposits. Their own funds as also borrowings from relatives and friends constitute the basis of their lending money.

3. Dealing in hundies: 'hundies' is a credit note of the indigenous bankers like Cheque for modern banks. These banks deal in the sale and purchase of various kinds of Hundies & are written in the local language.

2) Non-Banking Functions’,

Besides banking functions, indigenous bankers perform various types of non banking and trading functions as well This increases their income, though risk as well. Important non-banking functions are as under:

I. Speculative Trade: Indigenous bankers are known to be actively engaged in speculative trade. jiggery, wheat and other such products arc their field of operation.

2. Commission Agents: At many places. indigenous bankers also function as commission agents in the grain markets.

3. Other Business: With a view to recouping their losses on account of competition from the modern banks, the indigenous bankers undertake various non-banking activities such as trading and shop-keeping. Many indigenous bankers have become jewellers.

ECONOMY

Arthashastra means the science (sastra) of wealth/ earth/polity(artha)

What is the meaning of “Artha” ? Process of acquiring materialistic benefit for better livelihood – without ‘Greed’ ARTHASASTRA is the name of the text written by him. Kautilya is the author of the "Arthashastra". Popularly known as Chanakya, he authored "Arthashastra" in the Mauryan period.. He is widely known as “Chanakya” – son of Acharya Chanak, Acharaya Chanakya was Guru of Chandragupta Maurya, who built the Mauryan Empire - 300 BCE

This Arthasastra text covers

(i) State Administration

(ii) Economic Policy

(iii) Military Strategies :

The text incorporates Hindu philosophy includes ancient economic and cultural details on agriculture, mineralogy, mining and metals, animal husbandry, medicine, forests and wildlife

Topics in Arthashastra • Principles of Management • Management Education • Accounting systems • Corporate Governance • Time Management • Leadership Skills • Contracts • Selection of Employees • Consulting • Strategic Management • Handling competition • Expansion of Territory etc

Total of 15 volumes

1. Concerning Discipline
2. The Duties of Government Superintendents
3. Concerning Law
4. The Removal of Thorns
5. The Conduct of Courtiers
6. The Source of Sovereign States
7. The End of the Six-Fold Policy
8. Concerning Vices and Calamities
9. The Work of an Invader
10. Relating to War
11. The Conduct of Corporations

12. Concerning a Powerful Enemy
13. Strategic Means to Capture a Fortress
14. Secret Means
15. The Plan of a Treatise

History

- From 1st to 7th centuries India was estimated to have the largest economy in the word controlling about one third and one fourth of the world(timeline)
- The country was often referred as Swarnabhumi and Swarnadweep by many writers and travelers because of its prosperity
- During 18th century The British empire began to take roots in India, there by the Indian economic condition was slowly changed from being an exporter of processed goods to the exporter of raw materials and buyer of manufactured goods
- India begins to reindustrialize after independence, the process of rebuilding the Indian economy have been started
- As a part of this the first five-year plan was implemented in 1952
- From 1st to 7th centuries India was estimated to have the largest economy in the Due importance was given to establishment of modern industries, modern technological and scientific institutes, space and nuclear programs
- To overcome the problem of lack of capital, rise in population, huge expenditure on defense, inadequate infrastructure etc.
- India relied heavily on borrowing from foreign sources and finally agreed to economic liberalization in 1991
- The Indian economy is one of the fastest growing economies in the world today
- The high growth sectors have been identified which are likely to grow at a rapid pace
- The recent initiative of the Government of India such as Make in India, skill India, Digital India, Foreign policy 2015-20e.t.c is expected to help the economy in terms of export and import and trade balance

1. **Ancient Indian Economy**
2. **Medieval Indian Economy**
3. **British India's Economic System**
4. **Economy of Independent India**
5. **Current Indian Economy**
6.
   1. Ancient Time (776 – 300 BC)
   2. Medieval Time and Islamic Golden Age (5 – 15th Century)
   3. Age of Enlightenment or The Age of Reason (1770 – 1820)
   4. Industrial and Economic Revolution (1820 – 1929)
   5. War and Depressions (1929 – 1945)
   6. Post-War Economics (1945 –1970)
   7. Contemporary Economics (1970 – present)

The economy history of India since INDUS VALLEY civilization to 1700 AD can be categorised under this phase. During this Phase indian economy was very will developed. It has very good trade relation with other parts of world. Before the advent of the East India Company each village. Continue. in India was a self-sufficient entity and was economically independent as all the economies needs were fulfilled with in the village.

During the Mughal period (1526–1858 AD) India experienced unprecedeneted prosperity in history. The gross domestic product of India in the 16th century was estimated at about 25.1% of the world economy. An estimate of India's pre-colonial economy puts the annual revenue of Emperor Akbar's treasury in 1600 AD at £17.5 million (in contrast to the entire treasury of Great Britain two hundred years later in 1800 AD, which totalled £16 million). The gross domestic product of Mughal India in 1600 AD was estimated at about 24.3% the world economy, the second largest in the world. By this time the Mughal Empire had expanded to include almost 90 per cent of South Asia,

The arrival of East India Company in India caused a huge strain to the Indian economy and there was a two way depletion of resources.The British would buy raw materials from India at cheaper rates and finished foods were sold higher than normal price in Indian market. During this phase Indians share of world income declined from 22.3% to 3.8% in 1952

After India got independence from colonial rule in 1947, the process of rebuliding started various policies and schemes were formulated. 1st 5 years plan came in to implemention in 1952. there 5th year plan started by indian government, focused on the needs of the indian economy.

From 1951 to 1979, the economy grew at an average rate of about 3.1 percent a year in constant prices, or at an annual rate of 1.0 percent per capita. During this period, industry grew at an average rate of 4.5 percent a year, compared with an annual average of 3.0 percent for agriculture.
From FY 1980 to FY 1989, the economy grew at an annual rate of 5.5 percent, or 3.3 percent on a per capita basis. Industry grew at an annual rate of 6.6 percent and agriculture at a rate of 3.6 percent.
Liberalisation and its effects (1991 Onwards) : The emergence of India as a major economic power in the world happens to be one such idea". Since then economy has progressed immensely with GDP progressing at the rate of 6-8% per annum.
India's Economy is bound for slower growth. In recent months, Indian government has introduced Pro business economic reforms and outlined plans to increase. Spending on capital investment and large scale social programs. In the first three months of 2013 the GDP growth slowed to 4.8% and it is likely to go down further due to weak Consumption , Capital, investment & decline government spending.

Ancient times till 1707 AD	• Indus Valley Civilization (3500 BC - 1800 BC) • Economy relied on trade, facilitated by advancements in transport. • Traded terracotta pots, beads, metals, gemstones, and more. • Used ships to reach Mesopotamia for trade.
	• 6th Century BC - 3rd Century BC • Punch-marked silver coins minted during the Mahajanapadas era. • Increased trade activity and urban development.
	• Maurya Empire (321 BC - 185 BC) • Political unity and security led to a common economic system. • Increased agricultural productivity and trade. • Empire invested in building and maintaining roads throughout India. • Improved infrastructure, security, and standardized measurements boosted trade
	• 1st Century AD - 17th Century AD • India became the largest economy in the ancient and medieval world. • Controlled between 33% and 25% of the world's wealth.
Mughal Period (1526 AD - 1858 AD)	• Unprecedented prosperity under the Mughals. • India's GDP reached 25.1% of the world economy in the 16th century. • Uniform customs and tax system enforced. • Mughal Empire's GDP reached 24.3% of the world economy in 1600 AD (second largest). • Decline of the Mughals in the 18th century.
18th Century Onwards	• Rise of the Marathas and decline of the Mughal Empire. • British influence and decline of Indian industry.
British rule	• British East India Company used revenue to buy Indian goods and raw materials. • Colonial government used revenue for wars, not development. • India changed from exporting processed goods to exporting raw materials under British rule. • British colonial rule devastated India's economy. • India's share of world income declined from 27% to 3% under British rule.
India after independence	• 1950-1979 • India adopted centralized planning after independence in 1947. • Five Year Plans were implemented to develop the economy. • Investments were made in agriculture, infrastructure, and industries. • Economic growth was slow due to lack of capital, Cold War politics, and population rise. • From 1951 to 1979, the economy grew at 3.1% annually and industry grew at 4.5% annually. • Agricultural growth lagged behind industrial growth at 3.0% annually.
	• 1980-1990 • India's economy grew at 5.5% annually from 1980 to 1989.

	• Industry grew at 6.6% annually and agriculture at 3.6% annually. • Increased investment (from 19% to 25% of GDP) fueled economic growth. • India increasingly relied on foreign borrowing in the late 1980s. • This led to a balance of payments crisis in 1990, prompting economic reforms. • Liberalisation and its effects (1991 Onwards
	• Since 1991, • India's economy grew rapidly at 6-8% annually. • GDP went from $267 billion in 1992 to $1.85 trillion in 2012. • India became the world's third largest economy and a major foreign direct investment destination. • India's major industries include IT, telecommunications, textiles, and others. • Agriculture is still important, with rice, wheat, and cotton as major products. • The service sector is a leading contributor to the economy. • India has a strong network of science and technology institutions. • The government aims to increase manufacturing's share of GDP to 25% by 2025. • India is expected to become the world's second largest manufacturing economy by 2017. • India is one of the largest and fastest-growing markets for food and agriculture. • India is the world's third largest producer of food. • The Indian food industry is expected to grow significantly in the coming years. •